

On- versus off- balance sheet bank funding: Covenants as a signaling device

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Abstract

We consider a setting in which a bank seeks funding for two sequential projects: the first constrained by effort moral hazard, and the second by adverse selection. Allowing only for short-term debt contracts widely used in practice, we show that on-balance sheet funding strictly dominates off-balance sheet funding for all parameter values. Therefore, a rational, profit-maximizing bank will never choose the off-balance sheet funding mode in this setting. Our analysis provides a new rationale for the use of covenants in which covenants play a signaling role.

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1 Introduction

A bank has the choice to finance lending either on- or off- balance sheet. Segura and Zeng (2020) present a model to provide an economic rationale driven by information frictions for the emergence of off-balance sheet funding. Although they obtain an inefficient partial pooling equilibrium, they claim that off-balance sheet funding with voluntary support is a more effective signaling device than on-balance sheet funding. We reconsider their model, and show that an efficient fully separating equilibrium arises under on-balance sheet funding. Our result obtains even though we restrict attention to short-term debt contracts. We show that our on-balance sheet funding solution strictly dominates the off-balance sheet funding solution for all parameter values.

Segura and Zeng (2020) consider uninsured debt instruments, i.e., public bonds, but focus only on short-term straight debt contracts. Although we also restrict attention to short-term debt, we allow for debt features which are widely observed in practice in cases of asymmetric information. By allowing for the use of short-term debt with covenants, we show that banks would strictly prefer on- over off- balance sheet funding. This implies that the results in Segura and Zeng (2020) are driven by the assumption that only straight debt contracts are used. However, this assumption contradicts empirical evidence on debt contracts used in practice under asymmetric information which frequently feature covenants (see, e.g., Smith and Warner 1979, Billett et al. 2007, Chava and Roberts 2008, Nini et al. 2009, Roberts and Sufi 2009, Demiroglu and James 2010, and Christensen and Nikolaev 2012).¹

In the model, there are two investment opportunities for a bank – the first suffers from an effort moral hazard problem, while the second suffers from an adverse selection problem. The bank funds the first project on-balance sheet. Compared to off-balance sheet funding, on-balance sheet funding induces greater effort provision since it is associated with a lower face value of debt. The lower promised repayment associated with on-balance sheet funding relaxes the bank’s effort moral hazard constraint more so than with

¹The sample of loans considered in these papers may include debt contracts issued by both financial and non-financial firms. A quick scan of EDGAR confirms that banks do frequently rely on bonds for funding (see, e.g., Simion and Rigoni 2025), and these bonds frequently include covenants.

off-balance sheet funding. Regarding the project associated with adverse selection, we show there always exists a unique fully separating equilibrium in which only good banks invest. Separation arises through the violation of the bad bank's participation constraint rather than through the standard incentive compatibility constraint.

Considering identical informational frictions, because they allow only for short-term straight debt, Segura and Zeng (2020) suggest that under on-balance sheet funding, banks cannot obtain funding for the adverse selection project. Off-balance sheet funding allows a good bank to (partially) separate through the provision of voluntary support, but it leads to a lower effort level in the first project. Thus, while the provision of voluntary support relaxes the adverse selection constraint in the second project, it makes the moral hazard constraint in the first project more binding.

In summary, our proposed on-balance sheet solution strictly increases bank value compared to the off-balance sheet solution of Segura and Zeng (2020) for two reasons: First, in the off-balance sheet solution, good banks signal by burning money which leads to an inefficient partial pooling equilibrium in which some value-destroying bad banks invest. Second, since some bad banks (those who do not mimic the good banks) do not use the cash flows generated by the asset-in-place to repay the debt on the moral hazard project, the face value of debt for this project is higher which makes the effort moral hazard problem more binding and leads to lower effort provision. In contrast, signaling in our solution does not involve any value-destruction as we obtain a unique *fully* separating equilibrium in which only good banks invest and it does not negatively affect effort provision in the first project. In addition, our solution is more general than the one presented in Segura and Zeng (2020): the (partial) separation in their solution arises only when the bank's earnings from assets-in-place are sufficiently large, while in our solution, full separation arises even if cash flows from assets-in-place are zero. Finally, our on-balance sheet funding solution strictly dominates the off-balance sheet funding solution for any level of investment in the second project.

Further, not only does the on-balance sheet solution (with costless signaling) generate a greater aggregate surplus than the off-balance sheet solution, banks extract the full surplus while competitive investors in both periods make zero expected profits, in equi-

librium.² Hence, given the informational frictions in this model, our on-balance sheet funding mode results in strictly higher bank profits for all parameter values, implying that a rational profit-maximizing bank will always choose it over the off-balance sheet funding mode.

Like in Segura and Zeng (2020), we restrict our contract space to short-term debt contracts. However, we allow for the inclusion of contingencies in the debt contracts, in line with empirical evidence. Restricting the analysis to straight debt contracts is unjustifiable both on theoretical and empirical grounds: From a theory point of view, if allowed to do so, banks will choose debt contracts with contingencies because these contracts allow them to increase their value. From an empirical point of view, evidence suggests that under severe asymmetric information, the securities used are not straight debt contracts, but debt contracts that include other features (e.g., covenants).

Our analysis presents a new rationale for the inclusion of covenants. Traditional theories posit that covenants are designed to grant lenders control rights upon the realization of specific future states, and hence, reduce agency costs (e.g., Rajan and Winton 1995, Park 2000, Garleanu and Zwiebel 2009, and Donaldson et al. 2022). This would foster separation since higher quality borrowers would pick contracts with stricter covenants since they are less likely to violate the terms than lower quality borrowers. In contrast, our model highlights a distinct signaling mechanism. We have in mind affirmative or positive covenants, in particular drawdown covenants, where future drawdowns are contingent on achieving certain milestones. In our setting, this corresponds to investors providing a small initial amount upfront, with a commitment to fund the remaining investment only after verifying that the firm allocates the initial amount to the project. Both good and bad banks propose the same contract, but separation arises through different actions taken by good and bad banks after investors accept the proposed contract.

²Different from Segura and Zeng (2020), who assume that investors make zero expected profits, we prove that this is the case in equilibrium.

2 Model

2.1 Set-up

There are four dates $t \in \{0, 1, 2, 3\}$. There is a bank which has no funds of its own at $t = 0$ and obtains a pay-off from assets-in-place at $t = 1$. The bank has two investment opportunities, the first at $t = 0$ is of scale 1 unit and the second at $t = 2$ is of scale I units. Each project generates a payoff in the period following the investment. The funds are raised from competitive external investors. All agents are risk-neutral and the discount rate is 0.

The first project is always good (g), while the second project may be either good or bad (b). In either case, a project produces R per unit of investment if it succeeds and zero if it fails. A project succeeds with probability p_i . We assume that only good projects are profitable:

$$\mathbf{A1:} \quad p_g R > 1 > p_b R$$

The first project to be undertaken at $t = 0$ is subject to effort moral hazard. The project succeeds with probability p_g and fails with probability $1 - p_g$. The bank can exert unobservable effort $e \in [0, 1]$ at a cost $c(e)$ to reduce the probability of failure. Taking into account the bank's effort provision, the first project's success probability is $p_g + me$ and its failure probability is $1 - p_g - me$, with $m \leq 1 - p_g$ to ensure non-negative probability of failure. m is a constant which is interpreted as the marginal value of effort. The cost of effort $c(e)$ satisfies:

$$\mathbf{A2:} \quad c(0) = 0, c'(0) = 0, c'(e) > 0 \text{ and } c''(e) > 0 \text{ for all } e > 0, \text{ and } c'(1) > mR$$

The bank obtains pay-off Y at $t = 1$ from its assets-in-place. We assume that Y is not so large such that the $t = 0$ debt becomes riskless, as otherwise the first-best effort level could be implementable:

$$\mathbf{A3:} \quad Y < 1 - \alpha(p_g R - 1)$$

At $t = 1$, the bank privately learns whether the second project is good or bad. The second project to be undertaken at $t = 2$ is of type g with probability $\alpha \in (0, 1)$. The bank privately observes the project's type. We assume that α is small such that the average NPV of the pool is negative:

A4: $\alpha < \frac{1-p_b R}{(p_g-p_b)R}$

The timeline is illustrated in Figure 1.

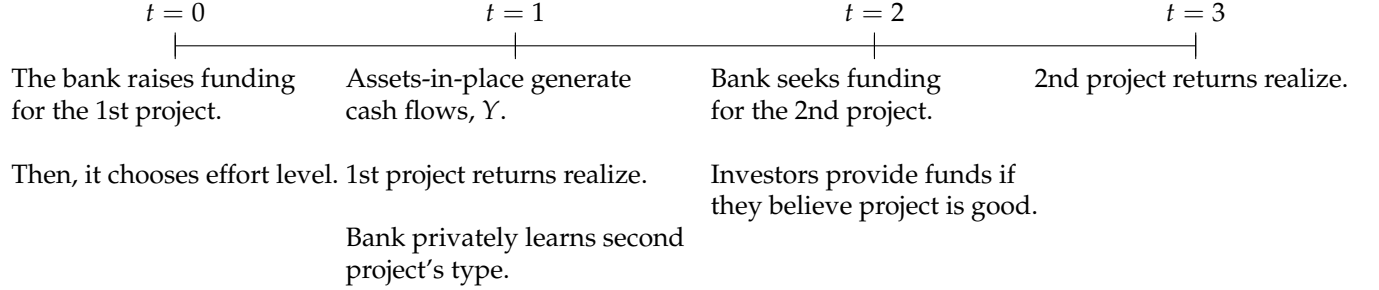


Figure 1: Timeline

2.2 The Game

We consider a two-stage sequential game, the second stage of which itself is a standard two-stage signaling game.

- Stage 1: At $t = 0$, competitive investors propose a contract, (D_1, Y) . The contract states that investors provide 1 unit of funding for the moral hazard project in return for repayment of D_1 if the project succeeds and Y if it fails.
- Stage 2a (which occurs at $t = 2$): the bank, now privately aware of its type, seeks to raise I units for the second project from competitive investors. The bank proposes a contract, (K, D_3) . The contract states that investors give an amount, K , to the bank at $t = 2$. The bank may use these funds to directly invest in the adverse selection project or consume it. In addition, investors obtain debt with face value, D_3 , in exchange for the remaining amount of funds necessary for undertaking the project which is either $I - K$ (if the bank invests K in the project) or I (if the bank consumes K).
- Stage 2b: given the contract offered by the bank in Stage 2a, investors will form beliefs about the second project's type. Given these beliefs, investors will decide

whether they will accept the contract and provide the funds necessary to undertake the project or not.

We look for the pure strategy Perfect Bayesian equilibria of this game that satisfy the Intuitive Criterion of Cho and Kreps (1987).³

2.3 First-best

In the first-best, the effort level exerted by the bank is observable and verifiable, and the type of the second project is publicly known. In this case, the bank always undertakes the first project and, given Assumption 2, the level of effort is:

$$c'(e^{FB}) = mR \quad (1)$$

The bank invests in the second project only if it is good.

2.4 Equilibrium

The two key dates of strategic interaction between the bank and investors are $t = 0$ and $t = 2$ when the bank raises funds for the projects. We solve for the equilibrium using backward induction.

2.4.1 Funding at $t = 2$

Below we list all the candidate equilibria of the subgame which starts at $t = 2$.

1. A candidate separating equilibrium in which only bad banks obtain funding.
2. A candidate pooling equilibrium in which both bank-types obtain funding.
3. A candidate pooling equilibrium in which neither bank-type obtains funding (market breakdown).

³The appropriate solution concept is Perfect Bayesian since in the second stage of the game banks may signal about their type through the use of K , which implies that inferences are made.

4. A candidate separating equilibrium in which both good and bad banks obtain funding, but at different terms.
5. A candidate separating equilibrium in which only good banks obtain funding.

Below we consider each candidate equilibrium.

Lemma 1 *There cannot exist a separating equilibrium in which only bad banks obtain funding or both good and bad banks obtain funding but at different terms.*

Proof. Due to limited liability, $D_3 \leq RI$. Suppose that in a separating equilibrium in which bad banks obtain funding without pooling with good banks, bad banks offer debt with face value $D_{3,b}$. Investors will participate only if the promised repayment satisfies $p_b D_{3,b} - I \geq 0$. However, even for the maximum feasible face value of debt, $D_{3,b} = RI$, this participation constraint is violated given that Assumption A1 states $p_b R < 1$. Therefore, there cannot exist a separating equilibrium in which bad banks obtain funding.

■

Lemma 2 *There cannot exist a pooling equilibrium in which both good and bad banks obtain funding at identical terms.*

Proof. Due to limited liability, $D_3 \leq RI$. In a pooling equilibrium in which both good and bad banks obtain funding at identical terms, investors will participate only if the promised repayment satisfies $\alpha p_g D_3 + (1 - \alpha) p_b D_3 - I \geq 0$. However, even for the maximum feasible face value of debt, $D_3 = RI$, this participation constraint is violated given Assumption A4 which states that α is small such that the average NPV of the pool is negative. Therefore, there cannot exist a pooling equilibrium in which both good and bad banks obtain funding. ■

At Stage 2a, the bank chooses D_3 and K to signal its type. Bank i 's profit is denoted Π_i and its participation constraint is given by:

$$\Pi_i = p_i(RI - D_3) - K \geq 0 \quad (2)$$

We derive the slope of $\Pi_i = 0$ in the (K, D_3) space by totally differentiating Equation (2) with respect to D_3 and K :

$$\left. \frac{dD_3}{dK} \right|_{\Pi_i=0} = -\frac{1}{p_i} < 0 \quad (3)$$

Suppose that investors at $t = 2$ provide K to all banks and provide the additional $I - K$ to a bank only if they believe that the bank is good. Denoting their expected profit as γ , investors make zero expected profits if:

$$\gamma = \alpha(p_g D_3 - I) - (1 - \alpha)K \geq 0 \quad (4)$$

We derive the slope of $\gamma = 0$ in the (K, D_3) space by totally differentiating Equation (4) with respect to D_3 and K :

$$\left. \frac{dD_3}{dK} \right|_{\gamma=0} = \frac{1 - \alpha}{\alpha p_g} > 0 \quad (5)$$

In Figure 2, we plot the binding participation constraints of banks (red lines) and investors (the blue line) in the (K, D_3) space. $\Pi_i = 0$ has a negative slope and it is linear in the (K, D_3) space. $\Pi_i = 0$ implies that $D_3 = RI$ for $K = 0$ and $D_3 = RI - \frac{I}{p_i}$ for $K = I$. $\gamma = 0$ has a positive slope in the (K, D_3) space. For $K = 0$, $\gamma = 0$ implies that $D_3 = \frac{I}{p_g}$. A bank of type i is willing to undertake the project in the region below its participation constraint, while an investor is willing to provide funds in the region above the participation constraint.

Lemma 3 *There cannot exist an equilibrium in which neither bank-type obtains funding.*

Proof. We present a graphical proof and refer to Figure 2. Suppose that neither bank-type obtains funding at $t = 2$. Such an equilibrium can lie anywhere on 2 outside the shaded region, including along the bad bank's participation constraint bordering the shaded region. Along its participation constraint, a bad bank is indifferent between participating or not, so may stay in, which leads to no funding for either bank-type (see Lemmas 1 and 2). A good bank can profitably deviate from a market breakdown equilibrium by offering

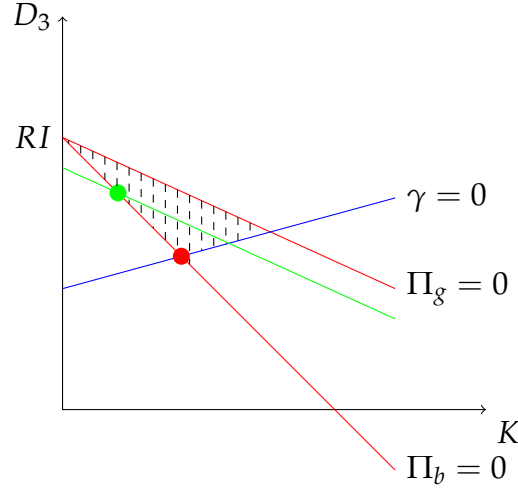


Figure 2: Funding at $t = 2$

a contract anywhere in the shaded region in Figure 2 above the bad bank's participation constraint and investors' participation constraint. Observing such a deviation, the application of the Intuitive Criterion implies that investors must attach a probability of 1 to the event that the offer comes from a good bank since bad banks' participation constraint is violated in this region. Investors are also strictly better off by accepting the offer since their participation constraint is slack in this region. Thus, the market breakdown equilibrium in which neither bank-type obtains funding does not survive the Intuitive Criterion.

■

Lemma 4 *There exists a separating equilibrium in which only good banks obtain funding.*

Proof. We present a graphical proof and refer to Figure 2. There exists a separating equilibrium with funding in the shaded region in Figure 2: here, investors' participation constraint is satisfied and they make (weakly) positive profits (i.e., $\gamma \geq 0$), a good bank's participation constraint is satisfied (i.e., $\Pi_g \geq 0$), while a bad bank's participation constraint is (weakly) violated (i.e., $\Pi_b \leq 0$). Funding is infeasible above the shaded region since the good bank's participation constraint is violated. Below the shaded region, the market breaks down since either the bad bank's participation constraint is satisfied and/or investors' participation constraint is violated.

Below we show that the unique equilibrium is at the intersection of Π_b and γ (the red dot) and only good banks obtain funding in this equilibrium. The red dot is characterized as follows:

$$K^* = \frac{\alpha p_b I (p_g R - 1)}{\alpha p_g + (1 - \alpha) p_b} \quad (6)$$

$$D_3^* = \frac{\alpha I + (1 - \alpha) p_b R I}{\alpha p_g + (1 - \alpha) p_b} \quad (7)$$

The lower contour of the shaded region is made up of the bad bank's participation constraint for $K \in [0, K^*]$ and investors' participation constraint for $K > K^*$. First, we rule out the separating equilibria anywhere in the shaded region above its lower contour.

An offer above the lower contour in the shaded region is not an equilibrium since the good bank can always profitably deviate by lowering the promised repayment D_3 by ϵ (where ϵ is strictly positive but arbitrarily small) for the same K . For $K \in [0, K^*]$, the constraint that binds first is the bad bank's participation constraint. In this case, investors' participation constraint is slack, so they make positive expected profits. For $K > K^*$, the constraint that binds first is investors' participation constraint.

Suppose that a separating equilibrium lies on the bad bank's participation constraint anywhere to the left of the red dot, say, at the green dot. To see why the green dot is not a stable equilibrium, consider the good bank's indifference curve going through the green dot (this is the green line running parallel to the $\Pi_g = 0$ line). A good banker can profitably deviate from this equilibrium by moving to a contract which lies in shaded triangle below the green line where the bad bank stays out since its participation constraint is violated. Observing the deviation, investors must hold the belief that the deviation comes from a good banker with probability 1 since the good banker is strictly better off and the bad banker is strictly worse off as a result of the deviation. Since investors' participation constraint is slack in this region, they accept the new offer and the conjectured equilibrium does not survive the Intuitive criterion. This argument rules out all equilibria on the bad bank's participation constraint to the left of the red dot.

Suppose that a separating equilibrium lies on investors' participation constraint to the right of the red dot. Again, a good bank can profitably deviate by making an offer lower

down on investors' participation constraint towards the red dot. While the good bank is strictly better off due to this deviation, the bad bank stays out since its participation constraint is violated in this region. Thus, such a deviation is interpreted by investors as one coming from good banks with probability 1, and since their participation constraint is satisfied, they accept the offer. Hence, the conjectured equilibrium does not survive the Intuitive criterion.

The analysis so far shows that no separating equilibrium except the allocation represented by the red dot can exist. Indeed, the red dot is an equilibrium since there is no profitable deviations in the shaded region for the good banker, who is strictly worse off moving anywhere in the shaded region. Thus, the red dot represents the unique equilibrium of this game. ■

Lemmas 1-4 rule out all possible equilibria other than the separating equilibrium at the red dot, which is established as a stable equilibrium. Hence, this is the only equilibrium of the game. We formally state this result in the proposition below:

Proposition 1 *There exists a separating equilibrium at (D_3^*, K^*) in which only good banks obtain funding and this is the unique equilibrium of the game played at Stage 2.*

2.4.2 Funding at $t = 0$

At $t = 0$, investors offer a contract to the bank which specifies the terms under which they will provide 1 unit to the bank for investment in the first project. Given the promised repayment, D_1 , investors' zero profit condition at $t = 0$ is as follows:

$$(p_g + me)D_1 + (1 - p_g - me)Y - 1 = 0 \quad (8)$$

From the zero profit condition, we derive the face value of debt, D_1 . The bank's expected profit is:

$$\Pi_1 = (p_g + me)(R - D_1 + Y) - c(e) \quad (9)$$

The bank maximizes its expected profit from the first project by choosing the level of effort. Taking the first order condition with respect to the effort level, solving the investor's zero profit condition (Equation (8)) for D_1 , and substituting it into the first order condition, we obtain the equilibrium effort level exerted by the bank:

$$c'(e^*) = m(R - D_1^* + Y) \quad (10)$$

$$= m \left(R - \frac{1 - Y}{p_g + me} \right) \quad (11)$$

Because $c(e)$ is strictly convex in e , $c'(e)$ is strictly increasing in e . Given Assumption A3, $D_1^* > Y$, implying that the debt is risky. In turn, this prevents equilibrium level of effort from reaching the first-best level, i.e., $e^* < e^{FB}$.

3 On- versus Off-balance sheet

Segura and Zeng (2020) consider a very similar setting and show that funding the first project off-balance sheet induces (partial) separation in the second project. Under the off-balance sheet financing mode, in the event of failure of the first project, there is no obligation for the bank to make repayments using cash flows arising from the bank's assets-in-place. However, the good bank can signal its second project's type by voluntarily using cash flows from the assets-in-place to repay the investors in the first project. The effort provision in the first project, e^{off} , is given by:

$$c'(e^{off}) = m \left(R - \frac{1 - \alpha Y}{p_g + me} \right) \quad (12)$$

The ratio of marginal cost of providing voluntary support to the marginal benefit is smaller for good banks compared to bad banks, implying that voluntary support can be an effective signaling device. There is a partial pooling equilibrium in which all good banks and a fraction of bad banks invest.

The on-balance sheet solution strictly dominates the off-balance sheet solution in Segura and Zeng (2020) for all parameter values:

1. Effort provision in the first project is higher under on-balance sheet funding compared to the off-balance sheet funding mode, i.e., $e^* > e^{off}$, since $\alpha < 1$. In the off-balance sheet solution, banks may use cash flows from their assets-in-place to signal their project's type. In contrast, in the on-balance sheet solution, the cash flows from a bank's assets-in-place are committed to the $t = 0$ investors which reduces the promised repayment, D_1 . This implies that the bank retains more of the surplus from exerting effort in this case, which leads to greater effort provision. Because the effort moral hazard constraint is binding, higher effort level implies higher bank value.
2. In relation to the second project, Segura and Zeng (2020) obtain a partial pooling equilibrium in which all good banks and a fraction of bad banks (say, $\pi \in (0, 1)$) invest if cash flows from the assets-in-place are sufficiently high or the level of the investment is sufficiently high; otherwise, there is a market breakdown equilibrium with no funding at all for either type. In contrast, our equilibrium is always separating in which only good banks invest even if the cash flows generated by the assets-in-place is 0. In our perfect separation solution, the value generated by the second project is $\alpha I(p_g R - 1)$, which is the probability that a bank is good multiplied by the NPV of a good bank's project. In contrast, if there is financing in the off-balance sheet solution of Segura and Zeng (2020), the value generated by the second project is $\alpha I(p_g R - 1) + \pi I(1 - \alpha)(p_b R - 1)$. The first term reflects the value generated by good banks. The second term reflects that a fraction of bad banks, π , from the $1 - \alpha$ probability mass, mimic good banks, resulting in value destruction given that $p_b R - 1 < 0$ (by Assumption A1). Since a partial pooling equilibrium entails $\pi > 0$, the second term is strictly negative for all parameters. Hence, the bank value under our solution strictly exceeds that under the off-balance sheet funding mode. Importantly, our solution is more general in the sense that it obtains for any value of cash flows arising from the assets-in-place (even if it is zero) and any strictly positive level of investment (even if it is ϵ), while the off-balance sheet funding solution of Segura and Zeng (2020) requires a sufficiently high (strictly positive) cash

flow from assets-in-place and/or a sufficiently high level of investment.

Proposition 2 *For all parameter values, on-balance sheet funding (when combined with costless signaling in the second project) strictly dominates off-balance sheet funding.*

In the choice of the funding mode, Segura and Zeng (2020) trade-off the efficiency gains from mitigating the adverse selection constraint with the efficiency losses from tightening the moral hazard constraint; mitigating one makes the other constraint more binding. In our case, the optimal contract at $t = 2$ delivers a fully separating equilibrium in the adverse selection project and it does so costlessly, so it does not negatively affect effort provision in the moral hazard project. As a corollary to Proposition 2, we state the following result:

Corollary 1 *A rational profit-maximizing bank will never choose the off-balance sheet funding mode.*

Consider a rational profit maximizing bank's choice of funding mode at $t = 0$. The on-balance sheet solution combined with optimal signaling in the second project leads to greater surplus creation since effort exertion in the first project is higher and there is no value destruction in the second project by bad banks who stay out due to the violation of their participation constraint. Additionally, note that banks extract the full surplus that is generated since investors in both periods make zero expected profits. Hence, given the informational frictions in this model, the off-balance sheet funding mode generates a lower profit for the bank compared to on-balance sheet funding. Therefore, a rational profit-maximizing bank will never choose the former.

4 Discussion

4.1 Empirical content

Segura and Zeng (2020) restrict their attention to straight debt. However, under severe asymmetric information conditions, such as those considered in this model, debt contracts frequently include contingencies in practice (see, for example, Smith and Warner

1979, Billett et al. 2007, Chava and Roberts 2008, Nini et al. 2009, Roberts and Sufi 2009, Demiroglu and James 2010, and Christensen and Nikolaev 2012). Hence, restricting attention to straight debt cannot be justified on empirical grounds. In our analysis, what allows separation in the second project is signaling through K – in equilibrium, investors give K units to all banks at $t = 2$ and provide the remaining $I - K$ units only to banks which commit the initial K units to investing in the project. This feature of the contract bears resemblance to contingent debt contracts which are used extensively in practice.

We have in mind affirmative or positive covenants, which require borrowers to take specific actions to ensure the continued availability of funds. In particular, one may think of drawdown covenants, where, after an initial borrowing by a bank, future drawdowns are contingent on achieving certain milestones. In this arrangement, the bank receives funds in tranches, with each tranche tied to meeting specific requirements. Chava and Roberts (2008) document that debt contracts often include provisions where the availability of ongoing funds and the terms at which these funds are provided are contingent on maintaining a minimum current ratio or net worth. In this case, if a firm's current ratio or net worth falls below some pre-specified thresholds – e.g., due to excessive dividend payments (in our setting, the bank consuming K rather than committing to invest it in the project) – then investors gain the control rights and may take steps to reduce the availability of funds and makes it more expensive. Indeed, Chava and Roberts (2008) find that following covenant violations, borrowers reduce investment, and especially so, in the case of severe information asymmetry. Hence, this evidence lines up well with our theory.

Also, Kaplan and Strömberg (2003) document the use of financial contracts in which venture capital firms, who usually fund projects suffering from severe information asymmetry (similar to the second project in our model), make future funding conditional on firms taking certain actions. In our setting, this is akin to suggesting that investors provide K and promise to pay $I - K$ upon observing whether or not the bank commits to investing K in the project. If K is consumed, further funding is discontinued.

4.2 Bankruptcy protocol

In this model, when the first project fails, the bank defaults on its payment to initial investors. However, it may still raise funds from new investors at $t = 2$ to finance the subsequent project subject to adverse selection. This process resembles Chapter 11 bankruptcy, as the bank maintains control over its assets, unlike in Chapter 7, where liquidation would prevent the bank from pursuing the second project. In fact, Chapter 11 bankruptcy emerges endogenously in our model. At $t = 1$, initial investors must decide whether to liquidate the bank – foregoing the opportunity for it to invest in the second project – or to reorganize, allowing it to continue operations. If they opt for liquidation, investors receive nothing. In contrast, reorganization enables good banks to generate a strictly positive surplus. The bank can always commit to paying initial investors an arbitrarily small but strictly positive amount, ϵ , which ensures that they are better off supporting reorganization over liquidation.

5 Conclusion

We present a model in which there are two investment opportunities for a bank; the first suffers from an effort moral hazard problem, while the second suffers from an adverse selection problem. We show that on-balance sheet funding strictly dominates off-balance sheet funding with voluntary support for all parameter values, even under weaker conditions on parameters. Our analysis implies that, given the informational frictions in the present setting, rational profit-maximizing banks do not choose the off-balance sheet funding mode even if they are restricted to using short-term contracts only. Hence, why voluntary support in off-balance sheet funding is observed remains an open question. It would be interesting to identify a different set of informational frictions such that this mode of financing arises as a best response by profit-maximizing banks.

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